William Welden

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Education

Trinity University San Antonio, TX
B.S. in Computer Science *Aug. 2021 – Dec. 2025*

Experience

Database Manager (Freelance) Jan. 2026 – Present
Pro Squares Janitorial Services

- Extended the company's TeamDesk operations database with custom tables and views for management reporting and workflows across 150 employees and 177 service locations
- Improved data accuracy by deduplicating and standardizing existing client and job records, establishing consistent formats for future entries

Open-Source Contributor Fall 2024
Blue Button Project

- Restored a long-failing test suite in an open-source XML–JSON library to fully passing by updating outdated dependencies and diagnosing a Windows line-ending (CRLF vs. LF) bug as part of a 3-person team
- Automated cross-platform CI, running 2 workflows (6 jobs across Ubuntu, Windows, and macOS) on every push and pull request, by building GitHub Actions pipelines for tests and coverage

Projects

DataTrader | *Go, HTMX, SQLite*

- Delivered a multi-user trading web app with authentication, persistent storage, and bulk CSV import, by building a REST-style Go server with an HTMX front end and SQLite

Go Interpreter | *Go*

- Built a working interpreter for a toy language, executing programs end to end, by implementing the lexer, parser, and evaluator from Thorsten Ball's *Writing an Interpreter in Go*

British | *Haskell*

- Delivered a complete language toolchain for an original esoteric language, by designing and implementing its lexer, parser, evaluator, and REPL in Haskell

TIPC Compiler | *C++*

- Compiled ILOC intermediate code to assembly, by implementing register allocation and instruction scheduling in C++

Keystroke | *TypeScript*

- Converted Markdown to HTML through a clean, extensible pipeline, by building a parser around an AST-based design in TypeScript

Technical Skills

Languages: Go, C#, Haskell, TypeScript, JavaScript, Java, Scala, C++, Python
Tools: GitHub, GitHub Actions, Bash, Google Cloud, Unity, Vercel, Convex, Clerk, TeamDesk